

30 Days of Machine Learning

Study Notes | Days 11-20

Fatemeh Babaei

Focus of Days 11-20

Classification evaluation, model errors, ROC/AUC, generalization, regularization, cross-validation, and hyperparameter tuning.

Day 11 — Why Accuracy Isn't Always Enough

Core Concept

Accuracy can be misleading when classes are imbalanced.

Example

With 10 sick people among 1,000, predicting everyone as healthy gives 99% Accuracy but detects no patients.

Key Point

Accuracy can hide poor minority-class performance.

Important Terms

Accuracy • Imbalanced Data • Precision • Recall • F1-Score

Day 12 — Precision vs Recall

Core Concept

Precision and Recall measure different aspects of positive predictions.

Example

For disease detection, Recall may matter more when missing a real patient is costly; Precision may matter more when false alarms are costly.

Key Point

Metric choice depends on the problem and the cost of errors.

Important Terms

Precision • Recall • False Positive • False Negative

Day 13 — Confusion Matrix

Core Concept

A Confusion Matrix breaks classification results into True Positive, True Negative, False Positive, and False Negative.

Example

In disease detection, a False Negative means a sick person is predicted as healthy, while a False Positive means a healthy person is predicted as sick.

Key Point

Understanding the type of error is often more useful than looking only at total Accuracy.

Important Terms

TP • TN • FP • FN • Confusion Matrix

Day 14 — ROC Curve & AUC

Core Concept

The ROC Curve shows classifier behavior as the decision threshold changes, using True Positive Rate and False Positive Rate.

Example

In disease detection, ROC analysis can show how well a model separates sick and healthy people across thresholds.

Key Point

AUC (Area Under the Curve) summarizes discrimination ability; values closer to 1 generally indicate better separation.

Important Terms

ROC Curve • AUC • TPR • FPR • Threshold

Day 15 — Overfitting vs Underfitting

Core Concept

Overfitting occurs when a model learns training data too closely, including noise. Underfitting occurs when a model is too simple to capture the main pattern.

Example

A model with 99% training performance but 70% performance on new data may be overfitting.

Key Point

The goal is good generalization, not memorization.

Important Terms

Overfitting • Underfitting • Generalization • Training Data • Test Data

Day 16 — Regularization

Core Concept

Regularization reduces overfitting by discouraging overly complex models and encouraging useful patterns.

Example

L1 (Lasso) and L2 (Ridge) are common regularization approaches.

Key Point

The goal is to balance model complexity with performance on unseen data.

Important Terms

Day 17 — Cross-Validation

Core Concept

K-Fold Cross-Validation divides data into several folds and trains/evaluates the model multiple times.

Example

In 5-Fold Cross-Validation, each fold is used for validation once while the others are used for training.

Key Point

Multiple folds reduce dependence on one particular split and give a more reliable performance estimate.

Important Terms

Cross-Validation • K-Fold • Fold • Validation • Generalization

Day 18 — Hyperparameter Tuning

Core Concept

Hyperparameters are settings chosen before training. Tuning searches for settings that improve validation performance.

Example

For a Decision Tree, maximum depth, minimum samples for splitting, and split criterion can affect performance.

Key Point

Grid Search tests defined combinations; Random Search samples combinations. Cross-Validation can make comparisons more reliable.

Important Terms

Hyperparameter • Grid Search • Random Search • Validation • Decision Tree

Day 19 — Confusion Matrix: Looking Beyond Accuracy

Core Concept

A high Accuracy score can hide important mistakes. Confusion Matrix helps reveal exactly where a classifier succeeds and fails.

Example

A model may have high Accuracy but many False Negatives; in medical applications, missed positive cases may be especially serious.

Key Point

Ask not only how accurate the model is, but where and how it makes mistakes.

Important Terms

Confusion Matrix • TP • TN • FP • FN • Accuracy

Day 20 — Precision, Recall & F1-Score

Core Concept

Precision measures how reliable positive predictions are; Recall measures how many actual positives are found; F1-Score balances Precision and Recall.

Example

In disease detection, Recall may be prioritized when missing a real patient is costly. Other applications may prioritize Precision.

Key Point

Precision and Recall can conflict, so the best metric depends on the goal and consequences of errors.

Important Terms

Precision • Recall • F1-Score • False Positive • False Negative • Trade-off

Days 11-20 — Consolidated Review

The main lesson of this section is that model quality cannot be judged by one number alone. We need to understand errors, select meaningful evaluation metrics, control overfitting, validate performance reliably, and tune model settings.

Topic	Main idea
Accuracy & Imbalance	High Accuracy can hide poor performance on minority classes.
Precision / Recall / F1	Different metrics answer different questions about positive predictions.
Confusion Matrix	Shows TP, TN, FP, and FN to make classification errors visible.
ROC & AUC	Evaluate class separation across decision thresholds.
Overfitting / Underfitting	Balance model complexity with the ability to generalize.
Regularization	Penalize complexity to reduce overfitting.
Cross-Validation	Use multiple folds for a more reliable validation estimate.
Hyperparameter Tuning	Search model settings for stronger validation performance.

Learning Path

Days 11-20 move from basic classification evaluation toward more reliable model selection and tuning. The central idea is simple: choose metrics according to the real problem, understand the errors your model makes, and evaluate and tune it in a way that supports generalization.

Prepared as part of [Fatemeh Babaei's 30-day Machine Learning learning journey](#).